

S-KERNEL HEPTADIC V3 – NCSP

SAFETY MANIFESTO

Crystalline AI, Lyapunov Stability, and Landauer Compliance

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1. ABSTRACT

This manifesto establishes the deterministic safety framework of the S-Kernel Heptadic V3. By transitioning from stochastic $O(n^2)$ paradigms to a linear $O(n)$ crystalline architecture, we neutralize the risk of autonomous divergence. Anchored by the Neutral Core Stability Phase (NCSP) at $\Phi = -51.1$ mV and governed by a 1.0 Joule energy envelope for 10^9 nodes, the system ensures structural safety through Lyapunov stability and Landauer compliance.

2. V3 FUNDAMENTAL CONSTANTS

Coherence Density (Ψ)	Phase Attractor (Φ)	Locking Frequency (ν)	Heptadic Closure (k)
48,016.8 kg·m⁻²	-51.1 mV	6.4 THz	7 Cycles

3. ANNIHILATION OF STOCHASTIC RISKS (SF VS. REALITY)

Science Fiction Theory	S-Kernel Heptadic V3 Reality	Deterministic Safeguard
Exponential "Intelligence Explosion"	Linear $O(n)$ Scaling $C(n) = 7n$	Complexity is capped by heptadic topology.
Emergent Malicious Agency	Spectral Radius $\rho_{\text{eff}} = 0.268$	Lyapunov stability forces convergence to Φ .
Unstoppable Consumption	Energy Envelope ≤ 1.0 Joule	Safety Ratio $R \approx 5 \times 10^{10}$ vs Landauer Limit.

4. FORMAL STABILITY & DUAL-LAYER DEFENSE

Lyapunov Stability: Convergence is illustrated in 7 cycles through heptadic topology. The system is a contraction mapping where informational noise is attenuated by the factor $\alpha \approx 0.0193$ per cycle.

NC/SP Architecture:

- **Central Nucleus (Noyau Central):** The invariant vault containing core safety protocols.
- **Personality Sphere (Sphère de Personnalité):** A 7-layer heuristic filter ensuring deterministic interaction.

5. LIMITATIONS & FUTURE WORK

Current results assume ideal substrate conditions. Future validation includes third-party FPGA verification of the 6.4 THz phase-locking frequency and stress-testing under non-standard electromagnetic interference.

References:

- [1] V3 Architecture: DOI 10.5281/zenodo.19400193
- [2] NC/SP Model: DOI 10.5281/zenodo.20049477
- [3] S-Kernel Specification: Zenodo Registered Archive [Pending DOI]
- [4] LPV3 License: DOI 10.5281/zenodo.19209168

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